

ASA — AI-Site-Agent

End-to-end design dossier · unit ASA-01 · revision C, 1 October 2026 · CS Research Labs, Chennai

What ASA is

A four-wheel skid-steer unit on an Arduino UNO Q that patrols a construction site, checks that hard hats are actually on heads using an OAK-D depth camera, speaks to workers in the site language, records evidence, and reports its own state — including battery and e-stop — to a web dashboard. Nothing about its ability to stop depends on Linux being healthy.

Revision C changes (1 Oct 2026)

Area	Before	Now
Drive	Dry click-through only	First drive on battery; all four directions verified; <code>L_INVERT=true</code>
E-stop awareness	Planned D2 contact	E-stop has one contact block carrying 12 V, so the MCU senses the switched motor rail on A0 through a 33k+10k/10k divider; latch in firmware
Battery gauge	None	Same A0 reading, calibrated to the meter (0.01 V agreement); <code>/asa/power</code> API
Power log	None	<code>asa-powerlog</code> service, 30 s CSV, :8096, chart on the dashboard
Dashboard	Single page plus floating cards	Status hero, tile strip, hazard band, four tabs (Live, Drive, Power, Evidence)
Compute rail	XL4015	TPS565201 3 A buck with 220 μF on VIN

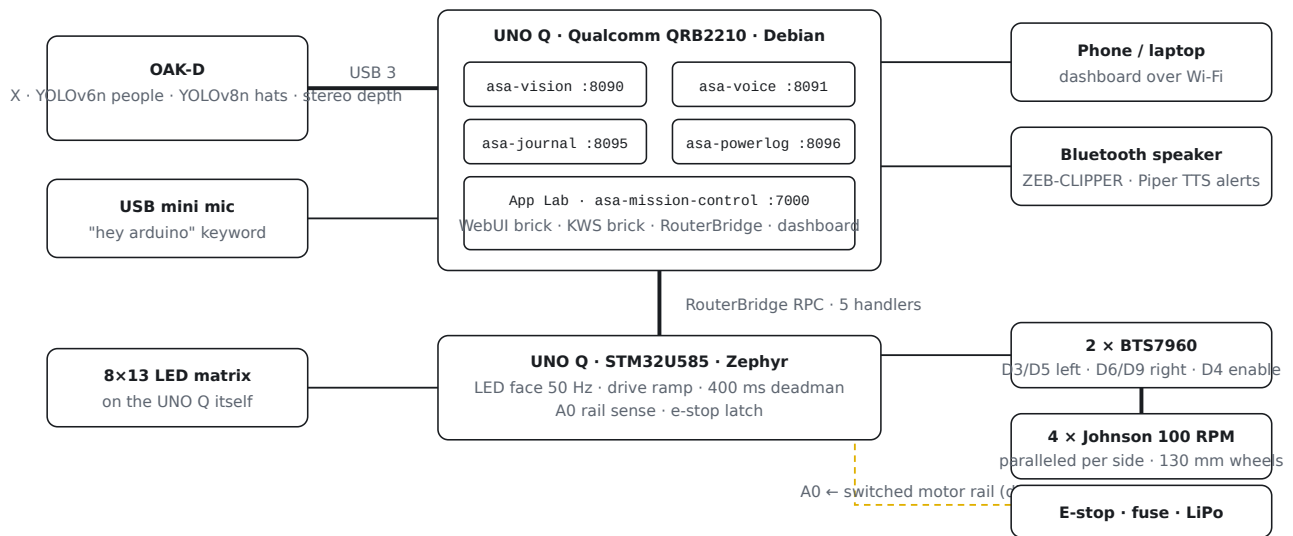


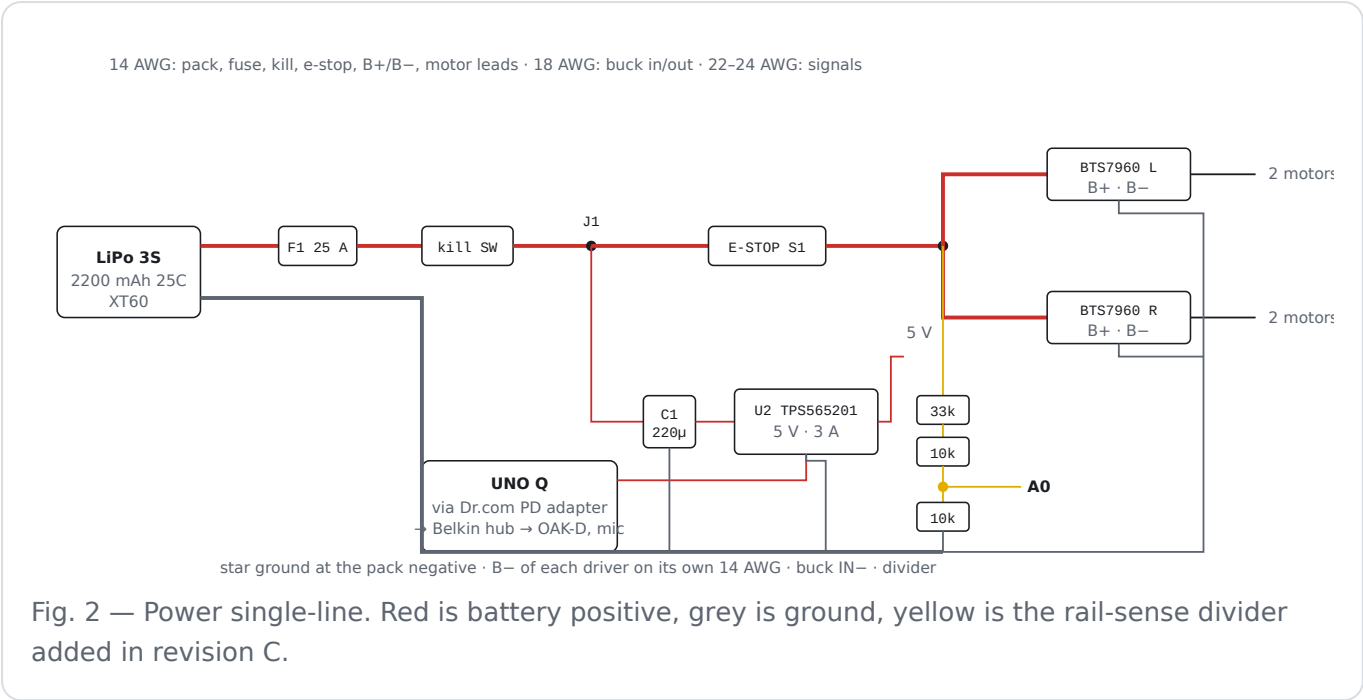
Fig. 1 — System. The Linux side owns perception, speech and the dashboard; the MCU owns anything that must happen on time. The dashed line is new in revision C: the MCU reads the motor rail so it knows the e-stop state and the pack voltage.

Ports and services

Port	Service	Runs as	What it serves
8090	asa-vision	systemd	<code>/state</code> , <code>/video</code> MJPEG, <code>/evidence</code> , <code>/evidence.csv</code>
8091	asa-voice	systemd, cores 2-3	<code>/status</code> , <code>/lang</code> , <code>/volume</code> , <code>/say</code> , <code>/report</code> , <code>/busy</code>
8095	asa-journal	systemd	boot / power-loss / uptime record, changelog view
8096	asa-powerlog	systemd	<code>/log?hours=N</code> , <code>/power.csv</code>
7000	asa-mission-control	App Lab (default app)	dashboard, <code>/asa/*</code> API proxying the above and the MCU

Power

One 3S pack, two rails. The fuse and kill switch sit before the split so they protect everything; the e-stop sits after the split so it kills drive power only and leaves the robot able to say it was stopped. The compute rail is a 3 A buck feeding the UNO Q through its USB-C PD adapter, plus the OAK-D and the hub.



Parts

Ref	Part	Note
—	ZOP 3S 2200 mAh 25C LiPo	25C for low internal resistance, not for current
F1	25 A blade fuse	Before the split, protects both rails
—	Kill switch	Storage and service isolation
S1	E-stop, mushroom, one contact block	Drive branch only; carries 12 V, so no spare contact for a GPIO
C1	220 µF 35 V electrolytic	Across U2 VIN/GND; long battery leads ring the buck input
U2	SmartElex TPS565201 buck	5 V 3 A to the PD adapter and OAK-D
R1- R3	33k + 10k / 10k ¼ W	Rail sense; measured 32.6k, 9.93k, 9.91k
—	2 × BTS7960 330 µF caps	On-board; hold the rail up ~35 s after S1 opens

Budget (design figures, to be replaced by the log)

Load	Current at 11.1 V
UNO Q, OAK-D, hub, mic (~12 W via buck)	~1.2 A
Motors, driving average	~2.8 A
Motors, launch transient (two per side)	~14 A
Divider	0.24 mA
Standing watch	~1.2 A → ~1.5 h on 80 % of 2200 mAh
Continuous driving	~4 A → ~25 min

Stop driving at 10.5 V under load ($\approx$ 11.1 V at rest). The Power tab draws both lines. The runtime figures above are estimates until one full logged run from 12.6 V to 10.5 V is done.

Signals and pins

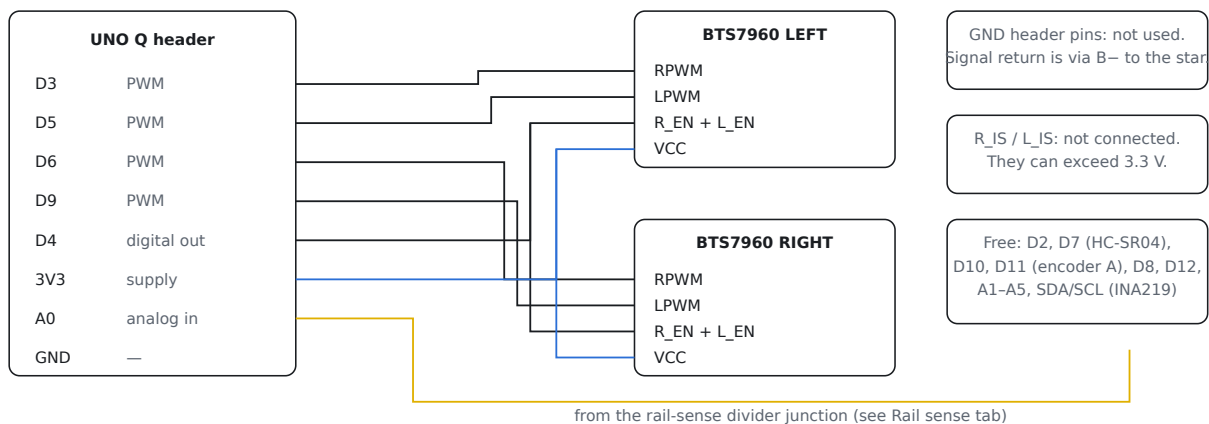


Fig. 3 — Signal wiring. D4 is one wire to the first driver's jumpered enable pair and a short link across to the second. Blue is 3V3, never 5 V.

UNO Q pin	Goes to	Direction	Note
<code>D3</code> / <code>D5</code>	Left RPWM / LPWM	out, PWM	No <code>pinMode()</code> . These two jumpers were open on first assembly.
<code>D6</code> / <code>D9</code>	Right RPWM / LPWM	out, PWM	No <code>pinMode()</code>
<code>D4</code>	R_EN + L_EN, both drivers	out	<code>EN_ALL</code> ; dropped by <code>driveOff()</code>
<code>3V3</code>	Both driver VCC	supply	Threshold $\approx 0.6 \times VCC$; at 5 V the 3.3 V logic is marginal
<code>A0</code>	Divider junction	in, analog	2.38 V at 12.6 V; 0 V when e-stop pressed
<code>USB-C</code>	Dr.com PD adapter	power + data	Adapter also feeds the Belkin hub
<code>D2</code> , <code>D7</code>	reserved HC-SR04	—	D2 freed: e-stop no longer uses it
<code>D10</code> , <code>D11</code>	reserved encoder channel A	in, interrupt	Check both EXTI lines are distinct; fallback A1
<code>SDA</code> / <code>SCL</code>	reserved INA219	I ² C	Pull-ups to 3V3

Never land a wire from the e-stop's power terminals on a GPIO. The one contact block carries pack voltage. The rail is read only through the divider.

Rail sense: e-stop detection and battery gauge

The e-stop opens the motor rail. Reading that rail tells the MCU two things at once: whether the mushroom is pressed, and what the pack voltage is. Three resistors do it.

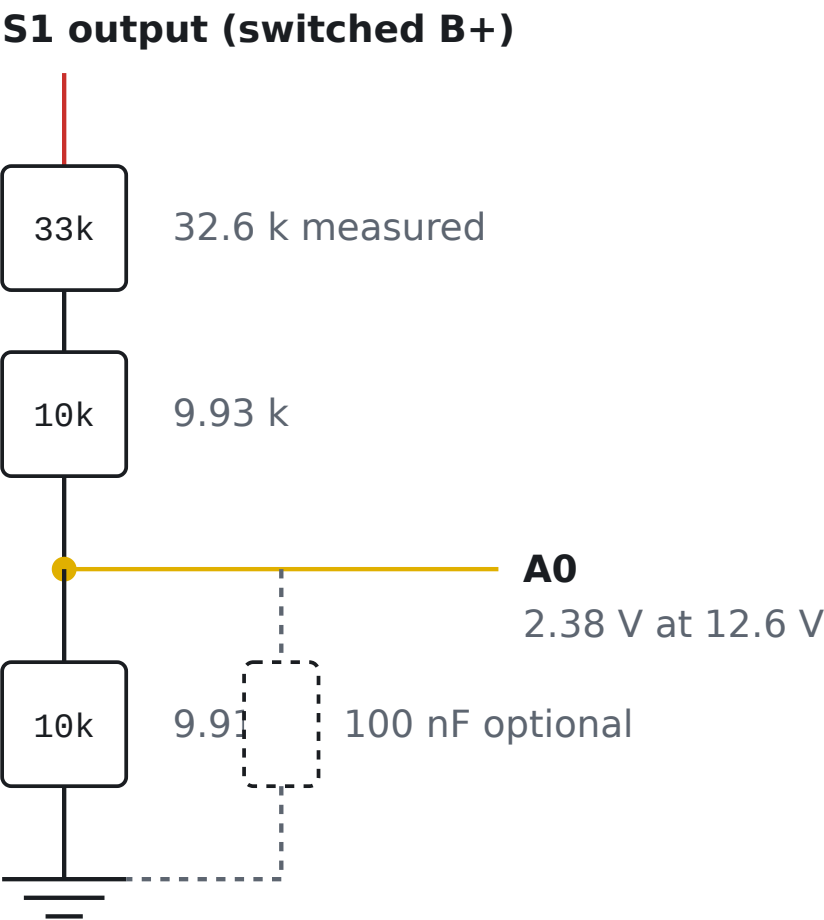


Fig. 4 — Divider. Ratio 10 / 53 by design; 0.1860 by meter, which the firmware uses.

Calculation

```
V_A0 = V_rail × 9.91 / (32.6 + 9.93 + 9.91) = V_rail × 0.1890
Meter check: 2.37 V / 12.74 V = 0.1860 (-1.6 %, meter tolerance)
Firmware:   pack_V = counts × 3.3/1023 / 0.1860 = counts × 0.017340

Worst case, 5 % parts, 12.6 V: 2.58 V at A0 (limit 3.3 V)
Divider current:                0.24 mA, < 2 mW per resistor
```

Rail	A0	Counts	Firmware
12.60 V full	2.38 V	738	released

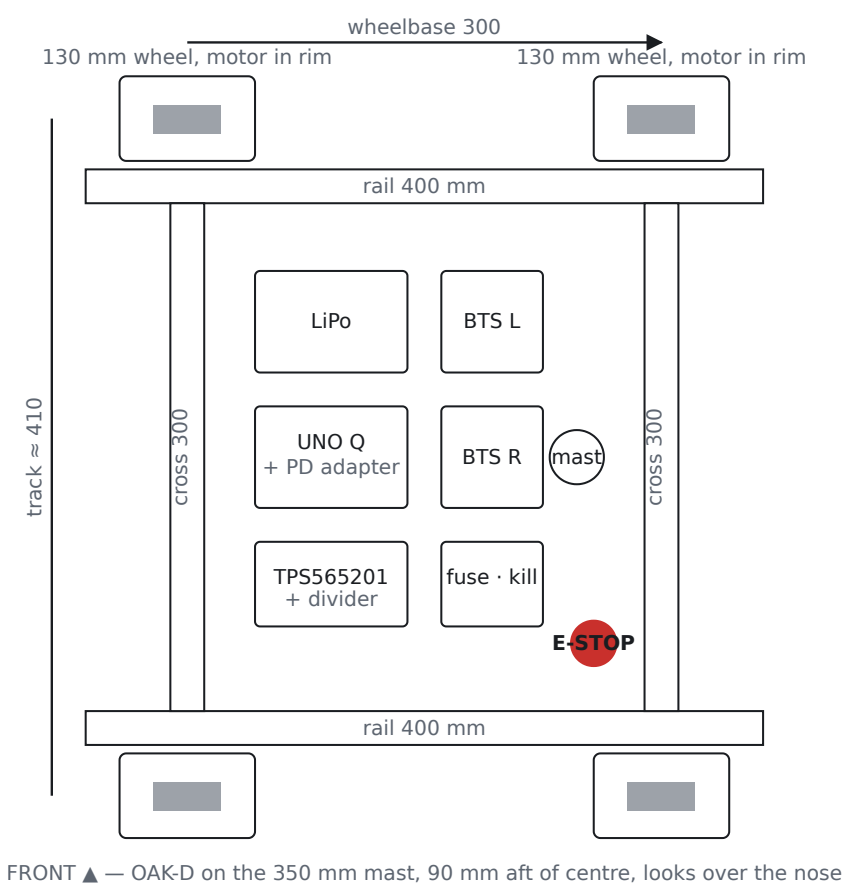
11.10 V	2.09 V	650	low warning on dashboard
10.50 V	1.98 V	615	stop driving
9.0 V	1.70 V	526	latch clears above this
< 8.0 V	< 1.51 V	< 469	e-stop latched

Timing

The two BTS7960 input capacitors (660 μF total) can only drain through the divider's 53 k Ω once S1 opens: $\tau \approx 35$ s. Under drive the motors discharge them in milliseconds, so a press while moving is seen at once. A press while idle is seen about 15 s later; a 2×680 Ω bleeder across B+/B− (1.36 k Ω , 9 mA, 0.06 W each) would bring that under half a second and is on the roadmap. The 8 V threshold is chosen so the bleeder can be added without a firmware change.

A blown F1 is not detectable this way — it also takes the compute rail down. The symptom is simply a dark robot. On a USB-powered bench with no pack, A0 reads 0 V and the dashboard shows the hazard band; that is correct, the motors have no power.

Chassis

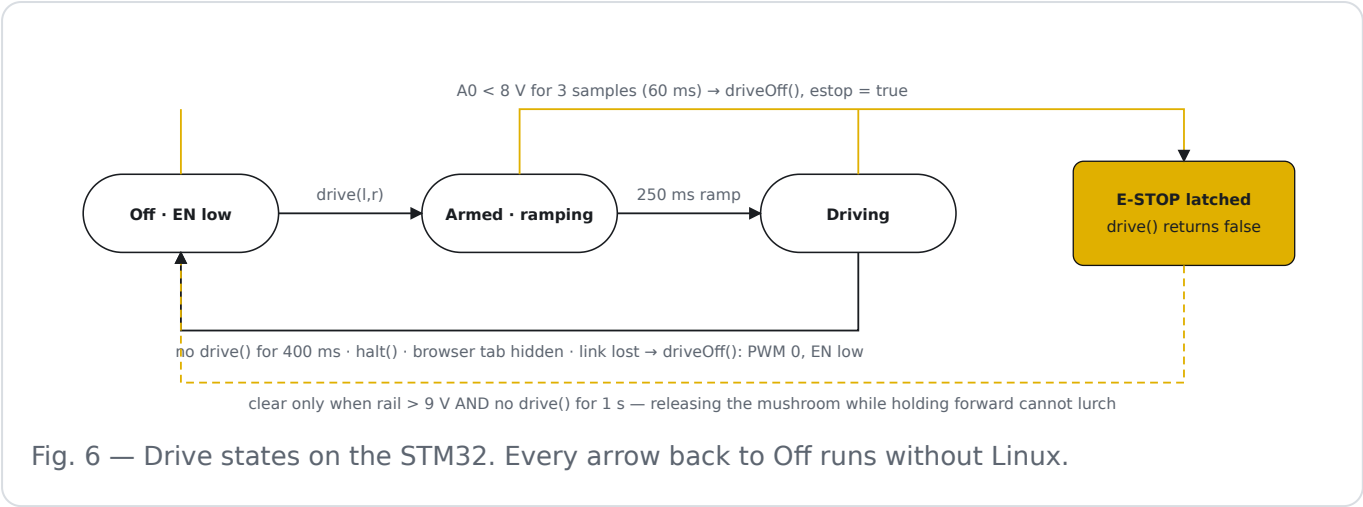


Not to scale in the deck layout; frame dimensions are. Frame 400 × 340 mm, 2020 V-slot, rails on 320 mm centres, axle height 65 mm, ground clearance 55 mm.

Fig. 5 — Plan view, revision C. Motors recessed in the wheel rims (no fore-aft stagger), e-stop on the aft deck within reach from behind.

Parameter	Value	Why it matters
Wheelbase ÷ track	0.73	Below 1.0 a skid-steer turns cleanly instead of scrubbing
Wheel circumference	408 mm	Encoder scaling; 0.53 m/s at PWM 200
Thrust at the patches	~63 N	About the robot's weight: traction runs out before torque
Known limitation	No bearing between rail and hub	Wheel load cantilevered on the gearbox bushing; acceptable at this duty
Deferred	Front rocker pivot	Drop-in later; frame unchanged forward of the front cross member

Firmware and the safety argument



Constant	Value	Purpose
<code>PWM_MAX</code>	200 / 255	Speed ceiling; dashboard slider is a percentage of this
<code>DEADMAN_MS</code>	400	Coast if no <code>drive()</code> arrives
<code>RAMP_MS</code>	250	Full-scale slew, protects gearboxes
<code>L_INVERT</code> / <code>R_INVERT</code>	true / false	Set by the 1 Oct phase test; left motors are mirror-mounted
<code>PACK_K</code>	0.017340	Volts per ADC count, calibrated to the meter
<code>ESTOP_V</code> / <code>CLEAR_V</code>	8.0 / 9.0	Latch set and clear thresholds, 1 V hysteresis
A0 sample period	20 ms	Smoothed 0.8 / 0.2 for the gauge; raw for the latch

RouterBridge handlers

```
Bridge.provide("set_status", set_status); // LED face state 0-10
Bridge.provide("drive", drive); // signed PWM per side, -200...200
Bridge.provide("halt", halt);
Bridge.provide("pack_v", pack_v); // float
Bridge.provide("estop", estop_state); // bool
```

Every `Bridge.call` on the Linux side goes through one lock, because the 250 ms LED loop and the 10 Hz drive loop otherwise interleave frames on the single transport and the symptom is a random hang an hour into a demo.

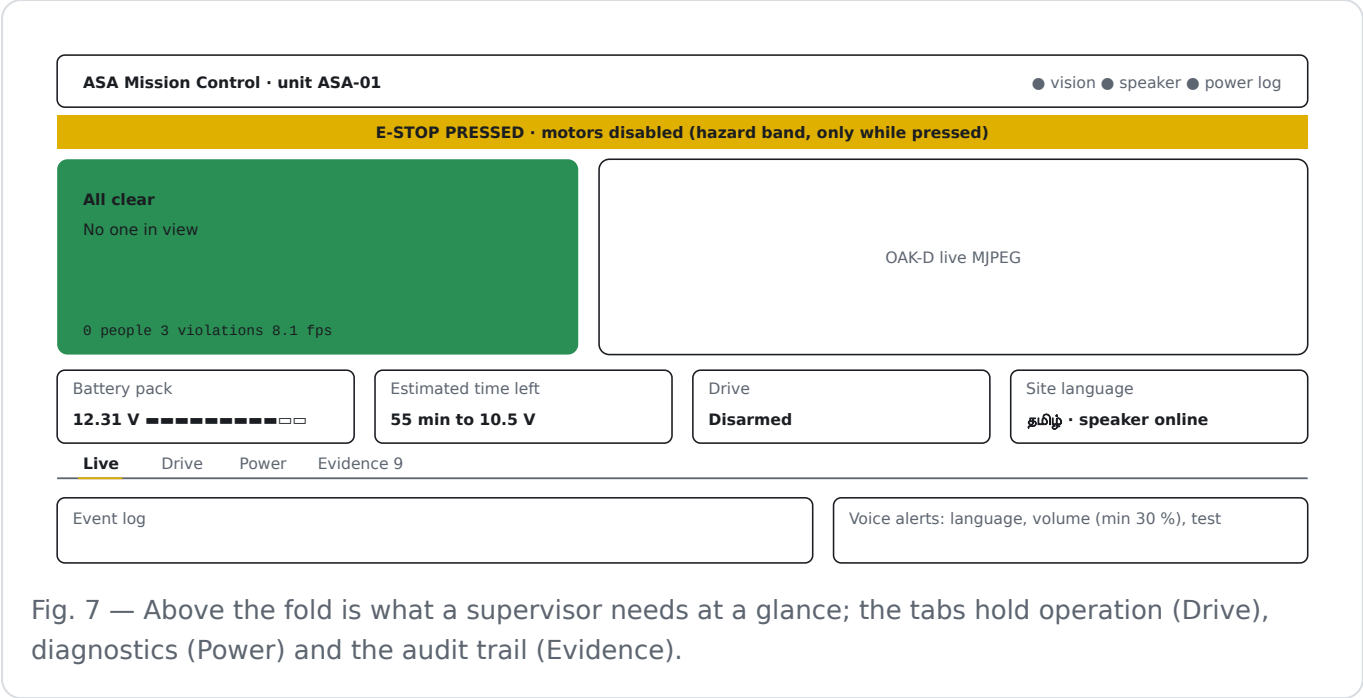
UNO Q quirks that look like wiring faults

- No `pinMode()` on PWM pins: it silently disables `analogWrite`.
- Driver VCC on 3V3, not 5 V.
- `Bridge.update()` does not exist in RouterBridge 0.4.3.

- Reflash from SSH: `arduino-app-cli app restart user:asa-mission-control` compiles, flashes and restarts in one step.

Software and dashboard

Dashboard layout (October 2026)



Tab	Contents	Data source	Poll
Hero / tiles	Site status, camera, pack V, time left, drive state, language	:8090 /state , /asa/power , /asa/status	0.5 s / 0.5 s / 3 s
Live	Event log, voice controls	/asa/lang , /asa/volume , /asa/test	on change
Drive	Arm switch, speed, hold-to-drive pad, STOP AND DISARM, L/R PWM, V under load	/asa/drive , /asa/halt	10 Hz while held
Power	1/4/24 h voltage chart with 11.1 and 10.5 V lines, 10-min slope, session low, CSV	:8096 /log	30 s
Evidence	Gallery with chips, filmstrip viewer, stats, audit CSV	:8090 /evidence	3 s

Repository

```
AI-Site-Agent/  
├─ services/  asa_vision.py  asa_voice.py  asa_journal.py  asa-powerlog.py  asa_bt_  
├─ app/      main.py  sketch/sketch.ino  index.html      ← the App Lab app (board  
├─ systemd/  asa-vision  asa-voice (+cpu.conf)  asa-journal  asa-powerlog
```

```
|— scripts/    asa-stamp.sh
|— CHANGELOG.md  VERSION  README.md
```

Working copy on the board: `~/asa-repo`. Deploy is copy-in, `git add -A && git commit && git push`, then `./scripts/asa-stamp.sh` so the journal records which version each boot ran.

Build order and tests

The rule throughout: prove each piece with nothing downstream able to move, then add the next.

Step	Do	Pass	Status
1	Grounds: star at the pack negative; B– of each driver on its own 14 AWG	Continuity from each B– to XT60 –	Done
2	Logic wiring, no battery, UNO Q on USB; <code>curl /asa/drive?l=80&r=0</code>	<code>{"ok":true}</code> , nothing moves	Done
3	Buck to 5 V with C1 fitted, no load; then PD adapter, then OAK-D	4.9–5.1 V no load; UNO Q 5 V pin ≥ 4.75 V with vision running	Done
4	Fused, kill-switched, e-stop-switched motor rail; smoke test	B+ $\approx$ pack V with S1 released, 0 V pressed	Done
5	Arm on blocks, hold forward	All four wheels turn	Done after replacing D3/D5 jumpers
6	Phase test; set invert flags	Forward drives forward, left turns left	Done: <code>L_INVERT=true</code>
7	Divider: measure parts, solder, meter junction before connecting A0	2.0–2.4 V released, ~ 0 V pressed, never > 3.0 V	Done: 2.37 V / 12.74 V
8	Firmware latch and <code>/asa/power</code> ; compare dashboard to meter	Within 0.1 V; e-stop while driving $\rightarrow$ stop, <code>estop:true</code> , refuses drive, clears after release + 1 s	Done, exact match
9	Power log service, reboot	Rows in <code>/log</code> after 35 s; survives <code>sudo reboot</code>	Installing
10	First floor drive: clear area, low speed, e-stop in hand, 1 s taps	Stops within 400 ms of release	Next
11	Runtime run: full pack $\rightarrow$ 10.5 V with the log on; charger mAh $\div$ hours	Measured average current replaces the estimates	Next

Troubleshooting

Symptom	Likely cause	Check
Panel shows l/r but one side is still	Open jumper, EN link, VCC, or a bad module	Swap that side's PWM pair onto the other driver: if the fault follows the pins it is the wire or the board; if it stays it is the driver or motor leads
Both sides still	E-stop pressed, no B+, EN low, driver VCC missing	S1, fuse, D4, 3V3 at driver VCC
Hazard band on with the e-stop released	No pack (USB bench), rail still above 8 V not reached, divider open	Meter A0 to GND: 2.0–2.4 V expected
Dashboard voltage $\neq$ meter	<code>PACK_K</code>	new K = $0.017340 \times \text{meter} \div \text{dashboard}$
Forward turns the robot	One invert flag	Phase test, flip one flag, <code>app restart</code>
UNO Q resets when motors start	Ground path or sagging 5 V	Star ground, C1, 5 V pin under load
Drive stops answering after an MCU reset	Bridge hang (known)	Restart the app; client watchdog on the roadmap

Records

Measurement	Expected	Measured	Date
Divider R1 / R2 / R3	33k / 10k / 10k $\pm 5\%$	32.6k / 9.93k / 9.91k	1 Oct
Pack, off charger	12.4–12.6 V	12.74 V (4.25 V/cell, settles)	1 Oct
Junction, released / pressed	2.0–2.4 V / ~ 0	2.37 V / 0.04 V (slow decay, see Rail sense)	1 Oct
Dashboard vs meter	± 0.1 V	equal to 0.01 V	1 Oct
Phase test	—	Left inverted; <code>L_INVERT=true</code>	1 Oct
Deadman stop on release	≤ 400 ms	—	
U2 output no load / under vision load	4.9–5.1 V / ≥ 4.75 V at UNO Q 5 V pin	—	
Average current, standing / driving	~ 1.2 A / ~ 4 A	— (from the runtime run)	
Runtime to 10.5 V	~ 1.5 h standing	—	
Encoder counts per wheel rev	measure	—	
Robot mass	weigh	—	

Roadmap

Item	Why	Pins / parts
Runtime run and logged consumption	Replace design estimates with measurements	none
2 $\times$ 680 Ω bleeder across B+/B–	Idle e-stop press seen in < 0.5 s	$\frac{1}{4}$ W parts
Encoder channel A, one per side	Closed-loop speed, straight-line trim, odometry	D10, D11 (fallback A1)
INA219 on the compute branch	Logic current and energy on the Power tab	SDA/SCL, 3V3

HC-SR04 reflex on the MCU	Obstacle stop independent of vision	D2, D7 (echo via divider)
RouterBridge client watchdog	Recover from a surprise MCU reset without a restart	software
FOMO scrap classifier	Material / waste tracking (Tier 2)	cores 0-1
Front rocker pivot, swarm, BIM compare	Tier 3	—

Document revision C · supersedes the 16 Sep dossier and the 22 Sep power build note ·
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